

Backend Architecture & Cartesia AI Voice Engine Specification

Complete engineering blueprint for Inbound/Outbound AI Cold Calling, Cartesia Sonic Realtime Voice Agent, CRM Database Schemas, and REST/WebSocket API Endpoints.

DOCUMENT VERSION	VOICE PIPELINE	TELEPHONY ENGINE	PUBLISHED DATE
2.0 (Cartesia Engine)	Cartesia Sonic (~90ms)	Twilio / SIP / WebRTC	August 2026

1 System Architecture & Core Modules Overview

CallX AI is an enterprise-grade AI-powered Cold Calling and CRM Platform built for automated, human-like voice communication across inbound customer reception and outbound sales campaigns.

Key Production Modules:

- **Dashboard & Realtime Analytics:** Real-time KPI cards (Total Calls, Daily Volumes, Conversion Success Rates, Pending Follow-ups), task to-do lists, call outcome visual reports, and workspace customization.
- **CRM & Master Leads Directory:** Full contact management, qualification scores, geolocation filtering, internal multi-author notes, document attachments (Quotes, Drawings, Proposals), and custom color-coded tags.
- **Voice Agent & Realtime Telephony:** Ultra-low latency voice calling powered by **Cartesia Sonic TTS** and streaming speech recognition, supporting both single direct calls and multi-line batch campaigns (1 to 10 concurrent lines).
- **Email Follow-Up Automation:** Post-call dynamic templating with rich HTML variable injection (`{name}` , `{company}` , `{agent}` , `{date}`), automated sending, and delivery tracking.
- **AI Agent Configuration:** Voice persona builder with Cartesia Voice IDs, tone adjustments (Professional, Friendly, Direct, Casual), speech rate controls, outbound cold calling script builder, and 24/7 inbound reception with live human handoffs.

2 Cartesia AI Voice Engine & Realtime Pipeline

The voice pipeline delivers a sub-400ms end-to-end conversational roundtrip (Voice Activity Detection → Speech-to-Text → LLM Token Stream → Cartesia Sonic Voice Synthesis → Telephony Output).

```
+-----+
| REALTIME CONVERSATIONAL VOICE PIPELINE | +-----+
+-----+ [ Caller: Phone / Twilio ] | (Bidirectional Audio Stream
- μ-law 8kHz / PCM 16kHz) v [ Twilio Media Streams / SIP WebSocket Gateway ] | +-----> [ VAD & Deepgram
Nova-2 Streaming STT ] (Realtime Transcription) | | v | [ LLM Orchestrator (GPT-4o / Claude 3.5) ]
(Inbound/Outbound Scenario Prompt) | | (Streams Tokens as they are generated) | v | [ Cartesia Sonic
WebSocket API (wss://api.cartesia.ai/tts/websocket) ] | | (Ultra-low latency audio chunks generation
< 90ms) | v [ Audio Player & Streamer ] -----> [ Caller hears natural voice ] | (Interruption / Barge-
in Detection: Immediately cuts Cartesia playback if user speaks) +-----+
```

2.1. Inbound & Outbound Calling Logic

FEATURE	OUTBOUND COLD CALLING ENGINE	INBOUND RECEPTION & SUPPORT ENGINE
Trigger	Manual click, scheduled time, or batch campaign runner.	Incoming customer phone call to Twilio dedicated number.
Opening Flow	AI speaks first with configured <code>openingGreeting</code> and value pitch.	AI answers with company greeting, business hours check, and intent triage.
Qualification	Asks defined <code>qualifyingQuestions</code> to score lead (Hot/Warm/Cold).	Extracts customer problem/inquiry and searches knowledge base / FAQ.
Action on Interest	Sends customized follow-up email, creates calendar event, tags lead as Won/Qualified.	Resolves query, books callback, or executes live SIP warm transfer to human agent.

3 Database Architecture & Schemas (PostgreSQL)

The relational database schema is designed for PostgreSQL with optimized indexing and foreign key constraints.

3.1. Master Customers & Leads Table (`customers`)

COLUMN NAME	DATA TYPE	CONSTRAINTS / VALUES	DESCRIPTION
<code>id</code>	BIGSERIAL	PRIMARY KEY	Unique customer ID
<code>full_name</code>	VARCHAR(150)	NOT NULL	Contact Full Name
<code>company_name</code>	VARCHAR(150)	DEFAULT ''	Company / Organization Name
<code>job_title</code>	VARCHAR(100)	DEFAULT ''	Position / Title (CEO, PM, GC)
<code>email</code>	VARCHAR(255)	INDEX	Email address
<code>phone</code>	VARCHAR(30)	NOT NULL, INDEX	Primary phone number in E.164 format
<code>website</code>	VARCHAR(255)	DEFAULT ''	Company website URL
<code>address</code> , <code>city</code> , <code>state</code> , <code>country</code>	VARCHAR	NULLABLE	Geolocation and physical address
<code>company_type</code>	VARCHAR(50)	DEFAULT 'General'	GC, Developer, Trade, Startup, Agency
<code>lead_status</code>	VARCHAR(50)	DEFAULT 'New'	New, Contacted, Qualified, Won, Lost
<code>lead_priority</code>	VARCHAR(50)	DEFAULT 'Warm'	Hot, Warm, Cold
<code>lead_quality</code>	VARCHAR(50)	DEFAULT 'Good'	Excellent, Good, Average, Poor
<code>last_contact_result</code>	VARCHAR(100)	DEFAULT 'Interested'	Interested, Call back, Meeting booked, No answer
<code>next_follow_up_date</code>	TIMESTAMPTZ	NULLABLE	Scheduled date for next follow-up
<code>status</code>	VARCHAR(20)	DEFAULT 'Active'	Active / Deactive
<code>reason_for_contact</code>	TEXT	DEFAULT ''	Context and sales intent
<code>tags</code>	VARCHAR[]	DEFAULT '{}'	Array of assigned tag labels
<code>created_at</code> , <code>updated_at</code>	TIMESTAMPTZ	DEFAULT NOW()	Creation and modification timestamps

3.2. Customer Notes Table (customer_notes)

COLUMN NAME	DATA TYPE	DESCRIPTION
id	UUID / BIGSERIAL (PK)	Unique note identifier
customer_id	BIGINT (FK → customers.id)	References customer record (ON DELETE CASCADE)
content	TEXT (NOT NULL)	Note body text
author	VARCHAR(100)	Author identifier (e.g. 'Admin', 'AI Voice Agent')
created_at	TIMESTAMPTZ	Creation timestamp

3.3. Customer Documents Table (customer_documents)

COLUMN NAME	DATA TYPE	DESCRIPTION
id	UUID / BIGSERIAL (PK)	Unique document identifier
customer_id	BIGINT (FK → customers.id)	References customer record (ON DELETE CASCADE)
name	VARCHAR(255)	Original file name (e.g. Proposal_v2.pdf)
size	VARCHAR(50)	File size display string (e.g. 3.2 MB)
type	VARCHAR(50)	Quote, Drawing, Proposal, Brief, Invoice
file_url	TEXT (NOT NULL)	S3 / Cloudflare R2 bucket download URL
upload_date	TIMESTAMPTZ	Upload timestamp

3.4. Call Logs & Transcripts Table (`call_logs`)

COLUMN NAME	DATA TYPE	DESCRIPTION
<code>id</code>	UUID / BIGSERIAL (PK)	Unique call log ID
<code>customer_id</code>	BIGINT (FK, Nullable)	Associated customer reference
<code>direction</code>	VARCHAR(20)	'inbound' or 'outbound'
<code>phone</code>	VARCHAR(30)	Destination or caller phone number
<code>status</code>	VARCHAR(50)	Completed, Failed, Queued, Upcoming, Callback Requested
<code>duration_seconds</code>	INTEGER	Call duration in seconds
<code>assignee</code>	VARCHAR(100)	AI Agent Voice Persona Name (e.g. 'Emma (Cartesia Sonic)')
<code>call_date</code> , <code>call_time</code>	DATE / TIME	Date and start time of call
<code>recording_url</code>	TEXT (Nullable)	Audio recording URL (S3)
<code>transcript</code>	JSONB (Nullable)	Turn-by-turn conversational transcript array with timestamps
<code>call_summary</code>	TEXT	AI-generated call summary and action items
<code>outcome_status</code>	VARCHAR(50)	Closed/Won, Qualified, Callback Requested, No Answer, Lost

3.5. AI Scenarios & Voice Settings Table (ai_scenarios)

COLUMN NAME	DATA TYPE	DESCRIPTION
id	UUID / VARCHAR(50) (PK)	Unique scenario ID
name	VARCHAR(150)	Scenario title (e.g. Outbound General Contractor Pitch)
category	VARCHAR(100)	Sales & Outreach, Support, Follow-Up
opening_greeting	TEXT	First spoken phrase when call connects
pitch_summary	TEXT	Core value proposition and product guidelines for LLM
qualifying_questions	JSONB (Array of Strings)	Array of questions AI must ask during call
action_on_interest	VARCHAR(255)	Automated workflow trigger (e.g. 'Send Proposal & Tag Hot')
cartesia_voice_id	VARCHAR(100)	Cartesia Voice Model ID (e.g. 'a0e99841-438c-4a64-b679-ae501e7d6091')
voice_speed	FLOAT	Voice playback speed multiplier (0.8x to 1.5x)
voice_tone	VARCHAR(50)	Professional, Friendly, Direct, Casual

3.6. Email Follow-up Templates Table (email_templates)

COLUMN NAME	DATA TYPE	DESCRIPTION
id	UUID / BIGSERIAL (PK)	Unique template ID
name	VARCHAR(150)	Template name (e.g. Proposal & Scope of Work)
category	VARCHAR(100)	Billing & Contracts, Follow-Up & Closing, Sales
subject	VARCHAR(255)	Email subject with placeholders (e.g. 'Proposal for {name}')
body_html	TEXT	HTML body supporting variables {name} , {company} , {agent}

4 RESTful & WebSocket API Specification

4.1. Authentication & Session

POST /api/v1/auth/login

Authenticate administrator or agent and return JWT bearer token.

Request: { "email": "admin@callx.ai", "password": "your_secure_password" }

Response (200): { "token": "eyJhbGciOiAi... ", "user": { "id": 1, "name": "Admin User", "role": "admin" } }

4.2. Leads & Customer Management (CRM)

GET /api/v1/customers

Retrieve filtered, sorted, paginated leads directory.

Query: ?search=john&leadStatus=Qualified&leadPriority=Hot&city=Vancouver&page=1&limit=25

POST /api/v1/customers

Create a new customer lead record with qualification scores and tags.

GET /api/v1/customers/:id

Fetch full customer profile including notes, documents, and call history.

POST /api/v1/customers/:id/notes

Append a new internal or AI note to customer timeline.

POST /api/v1/customers/:id/documents

Upload attachment file (Multipart form-data: `file`, `type`, `name`).

4.3. Voice Calls & Batch AI Campaigns (Cartesia Engine)

POST /api/v1/calls/dispatch-single

Initiate an immediate or scheduled outbound AI cold call to a specific contact.

Request: { "customerId": 42, "scenarioId": "sc_01", "cartesiaVoiceId": "sonic-english-emma" }

Response (202): { "status": "Queued", "callSessionId": "cs_99412", "phone": "+16045550199" }

POST `/api/v1/calls/launch-batch-campaign`

Launch high-volume cold calling campaign across 1 to 10 concurrent lines.

Request: { "customerIds": [10, 11, 12, 13], "scenarioId": "sc_01", "concurrentLines": 5 }

WS `/ws/media-stream/:callSessionId`

Bidirectional Telephony & Cartesia Audio Stream Gateway (Twilio Media Stream ↔ Backend ↔ Cartesia Sonic).

4.4. Inbound Call Handling Webhook

POST `/api/v1/telephony/inbound-webhook`

Twilio Inbound Voice Webhook: Returns TwiML WebSocket Stream configuration.

Response (TwiML): `<Response><Connect><Stream url="wss://api.callx.ai/ws/media-stream/in_123"/></Connect></Response>`

4.5. Automated Email Follow-ups

POST `/api/v1/emails/send`

Dispatch single or bulk follow-up email rendered with customer variables.

5 Production Architecture & Recommended Tech Stack

Production Stack Specifications:

- **Backend Framework:** Node.js (NestJS / TypeScript) or Python (FastAPI / AsyncIO).
- **Database & ORM:** PostgreSQL 16+ with Prisma ORM / SQLAlchemy.
- **Job & Call Queue Management:** Redis + BullMQ (Handles concurrent outbound dialing 1-10 lines and retry queues).
- **Voice Synthesis (TTS):** Cartesia Sonic WebSocket API (Low-latency streaming < 90ms).
- **Speech-to-Text (STT):** Deepgram Nova-2 Streaming WebSocket (Ultra-fast speech transcription & VAD).
- **LLM Intelligence:** OpenAI GPT-4o / Anthropic Claude 3.5 Sonnet (Function calling for CRM updates and email actions).
- **Object Storage:** AWS S3 / Cloudflare R2 (Audio recordings & customer PDF documents).

5.1. Cartesia WebSocket Streaming Implementation (Node.js Example)

```
import Cartesia from "@cartesia/cartesia-js"; const cartesia = new Cartesia({ apiKey:
process.env.CARTESIA_API_KEY }); const ws = cartesia.tts.websocket({ container: "raw", encoding:
"pcm_mulaw", sampleRate: 8000 }); // Stream LLM tokens to Cartesia and send audio directly to Twilio
Media Stream async function streamResponseToTwilio(llmTokenStream, twilioStreamWs, streamSid) { const
response = await ws.send({ model_id: "sonic-english", voice: { mode: "id", id: "a0e99841-438c-4a64-
b679-ae501e7d6091" }, // Emma Voice ID transcript: llmTokenStream, language: "en" }); for await
(const chunk of response.events("media")) { // Send 8kHz mulaw audio packet to caller with zero
buffer delay twilioStreamWs.send(JSON.stringify({ event: "media", streamSid: streamSid, media: {
payload: Buffer.from(chunk.data).toString("base64") } })); } }
```